

Research Aim & Implementation Objectives

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OVERALL AIM

Develop and evaluate an **ensemble AI-based detection framework** for cache-based side-channel attacks in Linux kernels achieving **>99% accuracy**, **<1% false positive rate**, and **<2% system overhead** using optimized HPC analysis.

6 OBJECTIVES – ADDRESSING ALL 5 RESEARCH GAPS

OBJ 1

G1

HPC Feature Engineering

Extract 15–20 HPC events (cache, TLB, branch).
Apply statistical selection with process metadata and 50ms temporal

✓ Ranked feature set with discrimination power analysis

OBJ 2

G5

Multi-Attack Dataset

Implement Flush+Reload, Prime+Probe, Flush+Flush, Spectre-V1, Rowhammer. Generate 100,000+ labeled HPC trace samples.

✓ Labeled attack-benign dataset with ground truth

OBJ 3

G2

Ensemble ML Models

Train RF + SVM + CNN-LSTM with 10-fold cross-validation. Combine via weighted majority voting ensemble.

✓ Trained models with performance comparison

OBJ 4

G3

Cross-Architecture Eval

Test on Intel i7, AMD Ryzen 5000, ARM Pi 4. Identify architecture-agnostic HPC features via transfer learning.

✓ Cross-architecture performance matrix

OBJ 5

G4

Real-Time Kernel Module

Linux kernel module for HPC monitoring. Sliding window analysis (100ms/50ms overlap). End-to-end ML inference pipeline.

✓ Functional kernel module + benchmarks

OBJ 6

G2

Evasion Resistance Tests

Test slow-down attacks, camouflaged attacks, and adversarial perturbation. Measure detection degradation vs. baseline.

✓ Evasion resistance report + mitigation strategies